KINETIC MODEL FOR ELASTIC WAVE PROPAGATION IN BOUNDED MEDIA AND APPLICATION TO HIGH-FREQUENCY STRUCTURAL ACOUSTICS

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Talk abstract

Semiclassical analysis of strongly oscillating solutions of classical wave systems, such as the Navier equation of elastodynamics, shows that the associated elastic energy density satisfies a Liouville-type transport equation in phase space. Here this model is extended by considering energetic boundary and interface conditions consistent with the boundary and interface conditions imposed to the solutions of the underlying wave system in a bounded medium. They are given in the form of power flow reflection/transmission operators for the bicharacteristic curves impinging on a boundary or an interface. Specular-like transverse boundary reflections, diffuse reflections, or fluid-structure coupling may be treated as a particular case of the proposed model. Nodal/spectral discontinuous "Galerkin" finite element methods and Monte-Carlo methods are implemented to integrate the transport equations supplemented with the proposed boundary and interface conditions. An application to the transient dynamics of a beam truss is proposed.

Kinetic model for elastic wave propagation

Let us consider the following Cauchy problem for elastic waves propagation in $\mathcal{O}\times\mathbb{R}$:

$$\begin{aligned}\varrho\partial_t^2\mathbf{u}_\varepsilon &= \boldsymbol{\nabla}_\mathbf{x}\cdot(\mathbf{C}:\boldsymbol{\nabla}_\mathbf{x}\otimes\mathbf{u}_\varepsilon)\,,\\ \mathbf{u}_\varepsilon(\cdot,0)=\mathbf{u}_\varepsilon^0\,,&\quad \partial_t\mathbf{u}_\varepsilon(\cdot,0)=\mathbf{v}_\varepsilon^0\,,\end{aligned}\tag{1}$$

depending on a small scaling parameter ε which quantifies the rate of change of the initial data $\mathbf{u}_\varepsilon^0$ and $\mathbf{v}_\varepsilon^0$ with respect to the sizes of $\mathcal{O}\subset\mathbb{R}^d$ or the propagation/observation distances. Here $\mathbf{u}_\varepsilon$ is the small displacement field of the medium about a quasi-static equilibrium considered as the reference configuration; and $\mathbf{C}(\mathbf{x})$ is the fourth-order elasticity tensor of the materials occupying $\mathcal{O}$ of which density is denoted by $\varrho(\mathbf{x})$. Since high-frequency waves are generated by an initial vibrational energy oscillating at wavelengths $\varepsilon\ll 1$, the initial conditions $\boldsymbol{\nabla}_\mathbf{x}\otimes\mathbf{u}_\varepsilon^0$ and $\mathbf{v}_\varepsilon^0$ shall be considered as strongly ε-oscillating functions, i.e. such that $(|\varepsilon\boldsymbol{\nabla}_\mathbf{x}|\varphi_\varepsilon)^2$ remains at least locally integrable on $\mathcal{O}$ [1]. The plane waves $\mathbf{u}_\varepsilon^0(\mathbf{x})=\varepsilon\mathbf{A}(\mathbf{x})e^{\mathrm{i}\mathbf{k}\cdot\mathbf{x}/\varepsilon}$ and $\mathbf{v}_\varepsilon^0(\mathbf{x})=\mathbf{B}(\mathbf{x})e^{\mathrm{i}\mathbf{k}\cdot\mathbf{x}/\varepsilon}$, $\mathbf{k}\in\mathbb{R}^d$, typically fulfill this condition.

Ray method

The classical approach to solve this problem is the ray method, consisting in seeking a solution of Eq. (1) in the form:

$$\mathbf{u}_\varepsilon(\mathbf{x},t)\simeq e^{\mathrm{i}S(\mathbf{x},t)/\varepsilon}\sum_{k=0}^{\infty}\varepsilon^k\mathbf{U}_k(\mathbf{x},t)\,,\tag{2}$$

where the phase function $S(\mathbf{x},t)$ is either real in the WKBJ method, or complex using Gaussian beams superposition [2]. Plugging the ansatz (2) into Eq. (1) it is shown, adopting an Eulerian point of view, that the phase function satisfies an eikonal equation and that the density $|\mathbf{U}_0|^2$ satisfies a linear transport equation of which coefficients depend on the phase function. Let us introduce the acoustic tensor $\boldsymbol{\Gamma}(\mathbf{x},\mathbf{k})$ and dispersion matrix $\mathbf{H}(\mathbf{s},\boldsymbol{\xi})$ of the propagation medium as follows:

$$\begin{aligned}\boldsymbol{\Gamma}(\mathbf{x},\mathbf{k})\mathbf{U}&=\varrho(\mathbf{x})^{-1}(\mathbf{C}(\mathbf{x}):\mathbf{U}\otimes\mathbf{k})\mathbf{k}\,,\\ \mathbf{H}(\mathbf{s},\boldsymbol{\xi})&=\varrho(\mathbf{x})(\boldsymbol{\Gamma}(\mathbf{x},\mathbf{k})-\omega^2\mathbf{I}_n)\,,\end{aligned}\tag{3}$$

where $\mathbf{s}=(\mathbf{x},t)$, $\boldsymbol{\xi}=(\mathbf{k},\omega)$, $(\mathbf{s},\boldsymbol{\xi})\in T^*(\mathcal{O}\times\mathbb{R})\equiv\mathcal{O}\times\mathbb{R}_t\times\mathbb{R}^d_\mathbf{k}\times\mathbb{R}_\omega$, $\mathbf{U}\in\mathbb{R}^n$, and $\mathbf{I}_n$ is the identity matrix of $\mathbb{R}^n$. $\mathbf{H}$ does not depend on the scale ε, that is to say high-frequency wave propagation $\varepsilon\ll 1$ is considered in a slowly varying "low frequency" medium. The eikonal and 0-th order transport equations read:

$$\begin{aligned}\mathbf{H}(\mathbf{s},\boldsymbol{\nabla}_\mathbf{s}S)\mathbf{U}_0&=\mathbf{0}\,,\\ \boldsymbol{\nabla}_\mathbf{s}\cdot\Big(\mathbf{U}_0^\mathsf{T}\boldsymbol{\nabla}_{\boldsymbol{\xi}}\mathbf{H}(\mathbf{s},\boldsymbol{\nabla}_\mathbf{s}S)\mathbf{U}_0\Big)&=0\,,\end{aligned}$$

respectively. Adopting a Lagrangian point of view, the pair $(\mathbf{s},\boldsymbol{\nabla}_\mathbf{s}S)$ is the solution of the Hamiltonian system associated to the elastic wave equation, which consists in solving the eikonal equation by the method of characteristics. Thus introducing the Hamiltonian $\mathcal{H}=\det\mathbf{H}$, the usual properties of the acoustic tensor $\boldsymbol{\Gamma}$ are such that:

$$\begin{aligned}\mathcal{H}(\mathbf{s},\boldsymbol{\xi})&=\prod_{\alpha=1}^{n}\mathcal{H}_\alpha(\mathbf{s},\boldsymbol{\xi})\,,\\ \mathcal{H}_\alpha(\mathbf{s},\boldsymbol{\xi})&=\varrho(\mathbf{x})\left(\lambda_\alpha^2(\mathbf{x},\mathbf{k})-\omega^2\right)\,,\end{aligned}\tag{4}$$

where λ_α^2 stands for the α-th (positive) eigenvalue of $\boldsymbol{\Gamma}$ with $1\leq\alpha\leq n$. As for isotropic elasticity for instance,

$\mathbf{\Gamma}(\mathbf{x},\mathbf{k}) = \lambda_{\mathrm{P}}^2(\mathbf{x},\mathbf{k})\hat{\mathbf{k}}\otimes\hat{\mathbf{k}} + \lambda_{\mathrm{S}}^2(\mathbf{x},\mathbf{k})(\mathbf{I}_d - \hat{\mathbf{k}}\otimes\hat{\mathbf{k}})$ with $\hat{\mathbf{k}} = \mathbf{k}/|\mathbf{k}|$ and $\lambda_\alpha(\mathbf{x},\mathbf{k}) = c_\alpha(\mathbf{x})|\mathbf{k}|$ of multiplicity 1 if $\alpha = \mathrm{P}$ or 2 if $\alpha = \mathrm{S}$, c_{P} and c_{S} being the elastic compressional and shear wave velocities, respectively, such that $c_{\mathrm{S}} < c_{\mathrm{P}}$. The corresponding Hamiltonian equations for $1 \leq \alpha \leq n$ are:

$$\begin{aligned}\frac{\mathrm{d}\mathbf{s}}{\mathrm{d}\tau} &= \nabla_{\boldsymbol{\xi}}\mathcal{H}_\alpha(\mathbf{s}(\tau),\boldsymbol{\xi}(\tau))\,, \qquad \mathbf{s}(0) = \mathbf{s}_0\,,\\ \frac{\mathrm{d}\boldsymbol{\xi}}{\mathrm{d}\tau} &= -\nabla_{\mathbf{s}}\mathcal{H}_\alpha(\mathbf{s}(\tau),\boldsymbol{\xi}(\tau))\,, \quad \boldsymbol{\xi}(0) = \boldsymbol{\xi}_0 \neq \mathbf{0}\,,\end{aligned} \tag{5}$$

in $T^*(\mathcal{O}\times\mathbb{R}) \setminus \{(\mathbf{s},\boldsymbol{\xi});\, \boldsymbol{\xi} = \mathbf{0}\}$, with an initial condition $(\mathbf{s}_0,\boldsymbol{\xi}_0)$ satisfying $\mathcal{H}_\alpha(\mathbf{s}_0,\boldsymbol{\xi}_0) = 0$. The rays are defined as the projections on $\mathcal{O}\times\mathbb{R}_t$ of the bicharacteristic curves $\tau \mapsto (\mathbf{s}_\alpha(\tau),\boldsymbol{\xi}_\alpha(\tau))$ solving (5), that is $\tau \mapsto \mathbf{s}_\alpha(\tau)$. They are associated to polarizations, or modes α defined by the dispersion relations $\mathcal{H}_\alpha(\mathbf{s},\boldsymbol{\xi}) = 0$ derived from the eikonal equation.

Kinetic model

The ray method with a real phase has some major drawbacks from either an Eulerian point of view or a Lagrangian point of view. The non linearity of the eikonal equation does not allow to superpose different phases, as would be required with several monopoles for instance; one way to circumvent this difficulty is to consider viscosity solutions. From the Lagrangian point of view, ray tracing is no longer possible on the caustics, where the rays stack, because the amplitudes $\mathbf{U}_k$ rapidly increase in their neighbourhood, and even blow up on the rays themselves; Gaussian beams may be considered to circumvent this shortcoming [2]. More generally the ansatz (2) of the ray method is only one *a priori* particular construction of high-frequency solutions of wave equations. This approach also requires rather strong regularity assumptions for the initial conditions of the phase S and amplitude $\mathbf{U}_0$. The more recent works of Gérard *et al.* [1], Papanicolaou & Ryzhik [3] or Akian [4], among others, on the semi-classical analysis of wave systems have generalized this theory for weaker assumptions on their high-frequency solutions and the initial conditions. They show that the energy density associated to all oscillating solutions, resolved in the phase space position × wave vector, satisfies a Liouville-type transport equation. The main mathematical tool for the derivation of a transport equation from a wave equation is the Wigner transform, of which high-frequency limit $\varepsilon \to 0$, the so-called Wigner measure, captures the vibrational energy density in phase space. The advantage of this representation is that it clears all classical difficulties inherited from ray approaches, and it yields global propagation properties of the energy for weakened initial conditions. In return, the explicit knowledge of the phase is lost. The eikonal equation is replaced by the dependence of the Wigner measure vs. $\boldsymbol{\xi}$, which gives its propagation directions as obtained from the dispersion equation $\mathcal{H}(\mathbf{s},\boldsymbol{\xi}) = 0$. Considering the solutions $\tau \mapsto (\mathbf{s}(\tau),\boldsymbol{\xi}(\tau))$ of the system (5) as the paths in phase space of some energy "particles" with a density denoted by $W(\mathbf{s}(\tau),\boldsymbol{\xi}(\tau))$, the latter shall satisfy:

$$\frac{\mathrm{d}W}{\mathrm{d}\tau} = \{\mathcal{H}, W\} = 0 \tag{6}$$

where $\{f,g\} = \nabla_{\boldsymbol{\xi}}f\cdot\nabla_{\mathbf{s}}g - \nabla_{\mathbf{s}}f\cdot\nabla_{\boldsymbol{\xi}}g$ is the usual Poisson bracket. Eq. (6) is a Liouville equation, which is the expression of the conservation of W in phase space starting from the initial data $W(\mathbf{s}_0,\boldsymbol{\xi}_0)$. As the dispersion matrix $\mathbf{H}$ is independent of time, Eq. (5) yields $\frac{\mathrm{d}\omega}{\mathrm{d}\tau} = 0$ and W has on $\mathcal{X} = T^*(\mathcal{O}\times\mathbb{R}) \setminus \{\mathbf{k} = \mathbf{0}\}$ the form:

$$W = \sum_{\alpha=1}^{n} W_\alpha \delta_0(\mathcal{H}_\alpha)\,. \tag{7}$$

The Wigner transform of $\mathbf{u}_\varepsilon$ and its associated Wigner measure are used to link W with the elastic energy density associated to the oscillating solutions $\mathbf{u}_\varepsilon$ of (1) for $\varepsilon \to 0$.

Boundary conditions

Boundary conditions for quadratic quantities such as the vibrational energy density shall be constructed on the basis of the boundary conditions applied to the underlying displacement and stress fields, for example Dirichlet or Neumann boundary conditions, or interface jump conditions between substructures (assuming that the thickness of that interface is much smaller than the wavelength). They basically translate into reflection/transmission operators for the energy rays. The consideration of boundary conditions in kinetic models raises some significant theoretical difficulties related to the polarization and the possible conversion of elastic waves, and to the critical angles of incidence arising for either transmission or reflection problems. Indeed, the "energy" density W having its support in the sets $\mathcal{X}_\alpha = \{(\mathbf{s},\boldsymbol{\xi}) \in \mathcal{X};\, \mathcal{H}_\alpha(\mathbf{s},\boldsymbol{\xi}) = 0\}$, the following condition has to be satisfied on the boundary $\partial\mathcal{O}\times\mathbb{R}$ or an interface $\Sigma \subset \mathcal{O}\times\mathbb{R}$, oriented by its normal $(\mathbf{n}, n_t)$:

$$\mathcal{H}_\alpha(\mathbf{s},\boldsymbol{\xi}) = a_\alpha(\mathbf{s})k_n^2 + b_\alpha(\mathbf{s},\mathbf{k}')k_n + \mathcal{H}'_\alpha(\mathbf{s},\mathbf{k}',\omega) = 0\,,$$

extracting the normal componant $k_n = \mathbf{k}\cdot\hat{\mathbf{n}}$ of the wave vector, with $\mathbf{k}' = (\mathbf{I}_d - \hat{\mathbf{n}}\otimes\hat{\mathbf{n}})\mathbf{k}$ for its tangential component. Denoting by $\Delta_\alpha(\mathbf{s},\mathbf{k}',\omega)$ the discriminant of this second-order equation with respect to k_n with real coefficients, the cotangent bundle to the boundary $T^*\partial\mathcal{O}\times\mathbb{R}$ splits as $T^*\partial\mathcal{O}\times\mathbb{R} = \mathscr{H}_\alpha\cup\mathscr{E}_\alpha\cup\mathscr{G}_\alpha$ ($\forall\alpha$), where the hyperbolic $\mathscr{H}_\alpha$, elliptic $\mathscr{E}_\alpha$ and tangent $\mathscr{G}_\alpha$ diffraction regions on $\partial\mathcal{O}\times\mathbb{R}$ are defined by (see for example [4, 5]):

$$\begin{aligned}\mathscr{H}_\alpha &= \{(\mathbf{s},\mathbf{k}',\omega)\in T^*\partial\mathcal{O}\times\mathbb{R};\,\Delta_\alpha(\mathbf{s},\mathbf{k}',\omega)>0\}\,,\\ \mathscr{E}_\alpha &= \{(\mathbf{s},\mathbf{k}',\omega)\in T^*\partial\mathcal{O}\times\mathbb{R};\,\Delta_\alpha(\mathbf{s},\mathbf{k}',\omega)<0\}\,,\\ \mathscr{G}_\alpha &= \{(\mathbf{s},\mathbf{k}',\omega)\in T^*\partial\mathcal{O}\times\mathbb{R};\,\Delta_\alpha(\mathbf{s},\mathbf{k}',\omega)=0\}\,,\end{aligned}$$

respectively. The first one corresponds to the transverse rays for which k_n is real (below critical incidence), the second one corresponds to the totally reflected rays for which $k_n\in\mathbb{C}\setminus\mathbb{R}$ (above critical incidence), and the third one corresponds to the tangent rays for which $k_n = 0$ in the local frame of the tangent plane to the boundary at $\mathbf{s}$ (critical or tangential incidence). As for an interface Σ, these definitions have to be extended on both sides since the acoustic tensor, and thus its eigenvalues, are *a priori* different; $T^*\Sigma$ is then the union of all these regions [5], but the latter are not necessarily disjoint. Mode conversions $\alpha\leftrightarrow\beta$ in $\mathscr{H}_\alpha\cap\mathscr{H}_\beta$ or $\mathscr{H}_\alpha\cap\mathscr{E}_\beta$ are driven by the condition $\frac{\mathrm{d}\omega}{\mathrm{d}\tau} = 0$ derived from Eq. (5) which enforces the conservation of the Hamiltonian, that is $\lambda_\alpha(\mathbf{x}(\tau_0^-),\mathbf{k}(\tau_0^-)) = \lambda_\beta(\mathbf{x}(\tau_0^+),\mathbf{k}(\tau_0^+))$ on any discontinuity front Σ of the density W given by $\Sigma = \{(\mathbf{s},\boldsymbol{\xi})\in T^*(\mathcal{O}\times\mathbb{R});\,\mathcal{S}(\mathbf{s},\boldsymbol{\xi})=0\}$, where therefore $\tau_0>0$ is such that $\mathcal{S}(\mathbf{s}(\tau_0),\boldsymbol{\xi}(\tau_0))=0$. Besides, the Rankine-Hugoniot condition on $\Sigma\cap(\cup_{\alpha=1}^n\mathcal{X}_\alpha)$ for the transport equation (6) reads:

$$[\![\{\mathcal{H},\mathcal{S}\}W]\!] = 0\,, \tag{8}$$

where $[\![f]\!] = f(\tau_0^+) - f(\tau_0^-)$ stands for the jump of f on the front. This condition expresses the conservation of the normal energy flux on the discontinuity, but *a priori* it does not describe how the energy density is distributed among the different modes by the reflections and transmissions on an interface Σ for example. However solving a Riemann problem at that interface gives a decomposition of the normal energy flux into rays moving forward and backward from it, including the effects of reflection and transmission. Thus the power flow reflection and transmission coefficients for the transport problem may be derived and understood in terms of some particular Riemann solutions [6]. It may be observed in addition that since W_β is necessarily zero in a neighbourhood of $\mathscr{E}_\beta$ (away from $T^*\Sigma$) because its support is in $\mathcal{X}_\beta$, its trace on $T^*\Sigma$ is zero as well, thus $W_\beta\equiv 0$ by the mode conversion $\alpha\to\beta$ within $\mathscr{H}_\alpha\cap\mathscr{E}_\beta$. For slender structures such as thick beams or plates for example, the power flow reflection/transmission coefficients at boundaries or interfaces are then derived from the dispersion properties of the constitutive and balance equations for these systems [7]. For the Mindlin-Reissner-Uflyand's thick plate kinematic model, the energy density is split into three propagative modes $\alpha\in\{\mathrm{T},\mathrm{S},\mathrm{P}\}$ with $\lambda_\alpha(\mathbf{x},\mathbf{k}) = c_\alpha(\mathbf{x})|\mathbf{k}|$ such that $c_\mathrm{T}<c_\mathrm{S}<c_\mathrm{P}$. Polarization T corresponds to transverse shear and the polarizations S, P correspond to bending and in-plane vibrational energies in the mean surface of the plate. The normal power flow reflection/transmission coefficients at the junction of plates and at a fixed or a free boundary are derived for the hyperbolic regions and the hyperbolic-elliptic region as functions of the angle between the plates.

Numerical simulations

Numerical methods with low numerical dispersion and dissipation errors are needed to perform long-time simulations of the transport equation in order to exhibit its diffusion limit if any. Monte-Carlo methods [8] and discontinuous finite element methods [9] both have these properties. Monte-Carlo methods are easy to implement since they are based on a physical interpretation of the transport equations. They also allow to compute local solutions, a decisive advantage over energetic methods (such as finite elements) for large-scale computations. Their main drawback is their lack of versatility for complex geometries as typically encountered in structural dynamics and acoustics. Finite element methods are, on the contrary, much more flexible and can be applied to truly complex geometries. Among them, the discontinuous "Galerkin" (DG) finite element method has originally been introduced in order to compute neutronic transfers; it is therefore adapted to the integration of transport equations. In the DG method, boundary fluxes at the edges or faces of the elements maintain the consistency of the numerical solution with the continuous, non-discretized transport equations, which are otherwise discretized on each element without any continuity relation between elements. The numerical fluxes are constructed in order to get to satisfy the reflection/transmission conditions derived from the analysis of Riemann solutions for boundaries and interfaces where the propagation media are discontinuous [10].

As an illustration of the previous setting, consider an hexahedral beam truss constituted by 13 three-dimensional thick beams, the 13$^{\text{th}}$ beam linking two opposite vertices of the truss through its diagonal. All beams have the same cross-sections and are made from the same homogeneous, isotropic elastic material. The initial con-

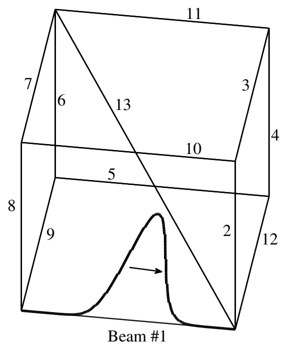

Figure 1: An hexahedral beam truss and the initial condition in beam $\#1$.

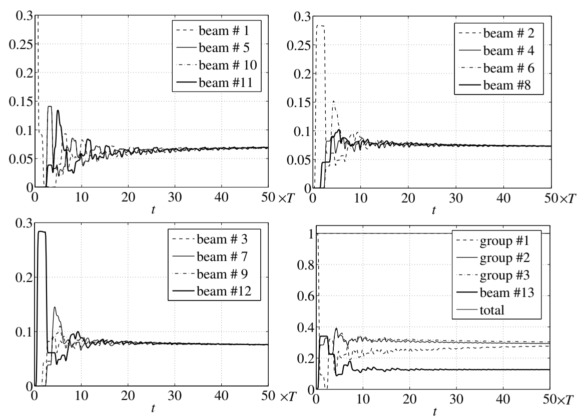

Figure 2: Evolution of the total vibrational energy for each beam of the truss.

dition is a pure compressional wave applied to, say, beam labelled $\#1$ and travels from the left to the right; see Figure 1. Figure 2 shows the evolution of the total vibrational energy for each beam computed by a DG method. The time scale $T = L/c_{\mathrm{T}}$ is the time needed by a transverse shear wave of velocity c_{T} to cross the edge of length L of the truss. The energetic behaviour of parallel beams is similar at late times; hence these results are displayed for each beam group, and then the sums of the energies for each group. The first group is constituted by the beams parallel to the initially loaded beam $\#1$ (beams $\#1, 5, 10, 11$), the second group is constituted by the vertical beams $\#2, 4, 6, 8$, and the third group is constituted by the remaining beams $\#3, 7, 9, 12$ whereas the diagonal beam $\#13$ is considered apart. The total energy computed for the entire truss is virtually conserved; this is a required property of the numerical scheme retained. Moreover, the vibrational energy in each beam tends to a limit as $t \to +\infty$ depending on its mechanical parameters. This phenomenom charaterizes the diffusive limit of transport equations holding at late times.

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